

2.21

Question: Registrants at a large convention are offered 6 sightseeing tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour planned by this convention?

Solution:

A person has 6 choices of tours for each of the 3 days. Therefore, the total number of ways is:

$$6 \times 6 \times 6 = 6^3 = 216$$

Answer: 216

2.22

Question: In a medical study, patients are classified in 8 ways according to whether they have blood type AB^+ , AB^- , A^+ , A^- , B^+ , B^- , O^+ , or O^- , and also according to whether their blood pressure is low, normal, or high. Find the number of ways in which a patient can be classified.

Solution:

There are 8 blood types and 3 blood pressure categories. Using the multiplication rule:

$$8 \times 3 = 24$$

Answer: 24

2.23

Question: If an experiment consists of throwing a die and then drawing a letter at random from the English alphabet, how many points are there in the sample space?

Solution:

A die has 6 faces, and the English alphabet has 26 letters. The total number of outcomes is:

$$6 \times 26 = 156$$

Answer: 156

2.24

Question: Students at a private liberal arts college are classified as being freshmen, sophomores, juniors, or seniors, and also according to whether they are male or female. Find the total number of possible classifications for the students of that college.

Solution:

There are 4 class levels (freshman, sophomore, junior, senior) and 2 genders (male, female). The total number of classifications is:

$$4 \times 2 = 8$$

Answer: 8



2.25

Question: A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

Solution:

For each of the 5 styles, there are 4 colors. Thus, the total number of pairs is:

$$5 \times 4 = 20$$

Answer: 20

2.26

Question: A California study concluded that following 7 simple health rules can extend a man's life by 11 years on average and a woman's life by 7 years. These 7 rules are as follows: no smoking, get regular exercise, use alcohol only in moderation, get 7 to 8 hours of sleep, maintain proper weight, eat breakfast, and do not eat between meals. In how many ways can a person adopt 5 of these rules to follow:

(a) if the person presently violates all 7 rules?

(b) if the person never drinks and always eats breakfast?

Solution:

(a) If the person violates all 7 rules, choosing 5 rules to follow is equivalent to choosing 5 items from 7. This is a combination problem:

$$\binom{7}{5} = \frac{7!}{5!(7-5)!} = \frac{7 \times 6}{2 \times 1} = 21$$

(b) If the person never drinks and always eats breakfast, these two rules are fixed. The person must choose 3 additional rules from the remaining 5:

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5 \times 4}{2 \times 1} = 10$$

Answers:

(a) 21

(b) 10

2.27

Question: A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?

Solution:

The total number of plans is the product of the choices:

$$4 \times 3 \times 2 \times 2 = 48$$

Answer:

2.28

Question: A drug for the relief of asthma can be purchased from 5 different manufacturers in liquid, tablet, or capsule form, all of which come in regular and extra strength. How many different ways can a doctor prescribe the drug for a patient suffering from asthma?

Solution:

There are 5 manufacturers, 3 forms (liquid, tablet, capsule), and 2 strengths (regular, extra). The total number of prescriptions is:

$$5 \times 3 \times 2 = 30$$

Answer:

2.29

Question: In a fuel economy study, each of 3 race cars is tested using 5 different brands of gasoline at 7 test sites located in different regions of the country. If 2 drivers are used in the study, and test runs are made once under each distinct set of conditions, how many test runs are needed?

Solution:

The total number of test runs is:

$$3 \times 5 \times 7 \times 2 = 210$$

Answer:

2.30

Question: In how many different ways can a true-false test consisting of 9 questions be answered?

Solution:

Each question has 2 possible answers (true or false). For 9 questions:

$$2^9 = 512$$

Answer:

2.31

Question: A witness to a hit-and-run accident told the police that the license number contained the letters RLH followed by 3 digits, the first of which was a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

Solution:

The first digit is fixed as 5. The second digit can be any of the remaining 9 digits (0–9 except 5), and the third digit can be any of the remaining 8 digits. Thus:

$$9 \times 8 = 72$$

Answer:

2.32

Question:

- (a) In how many ways can 6 people be lined up to get on a bus?
- (b) If 3 specific persons, among 6, insist on following each other, how many ways are possible?
- (c) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?

Solution:

- (a) The total number of arrangements of 6 people is:

$$6! = 720$$

- (b) Treat the 3 specific people as a single "block." There are $4!$ ways to arrange the block and the other 3 people, and $3!$ ways to arrange the people within the block:

$$4! \times 3! = 24 \times 6 = 144$$

- (c) First, calculate the total arrangements where the 2 specific people are together. Treat them as a single "block," giving $5!$ arrangements of the block and the other 4 people, and $2!$ arrangements within the block:

$$5! \times 2! = 120 \times 2 = 240$$

Subtract this from the total arrangements:

$$6! - 240 = 720 - 240 = 480$$

Answers:

(a)

(b)

(c)



2.33

Question:

- (a) In how many different ways can a student check off one answer to each question on a multiple-choice test consisting of 5 questions, each with 4 possible answers?
- (b) In how many ways can a student check off one answer to each question and get all the answers wrong?

Solution:

- (a) Each question has 4 choices. For 5 questions:

$$4^5 = 1024$$

- (b) To get all answers wrong, the student must avoid the correct answer for each question. For each question, there are 3 incorrect choices. Thus:

$$3^5 = 243$$

Answers:

- (a)
- (b)

2.34

Question:

- (a) How many distinct permutations can be made from the letters of the word COLUMNS?
- (b) How many of these permutations start with the letter M?

Solution:

- (a) The word "COLUMNS" has 7 unique letters. The total number of permutations is:

$$7! = 5040$$

- (b) Fix the first letter as M. The remaining 6 letters can be arranged in:

$$6! = 720$$

Answers:

- (a)
- (b)

2.35

Question: A contractor wishes to build 9 houses, each different in design. In how many ways can he place these houses on a street if 6 lots are on one side of the street and 3 lots are on the opposite side?

Solution:

First, choose 6 houses for one side of the street ($\binom{9}{6}$) and arrange them. Then arrange the remaining 3 houses on the other side:

$$\binom{9}{6} \times 6! \times 3! = 84 \times 720 \times 6 = 362,880$$

Answer:

2.36

Question:

- (a) How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 if each digit can be used only once?
(b) How many of these are odd numbers?
(c) How many are greater than 330?

Solution:

(a) The first digit cannot be 0. Choose the first digit from {1, 2, 3, 4, 5, 6} (6 choices). The second digit can be any of the remaining 6 digits, and the third digit can be any of the remaining 5 digits:

$$6 \times 6 \times 5 = 180$$

(b) For the number to be odd, the last digit must be {1, 3, 5}. Choose the last digit (3 choices), the first digit (5 choices, excluding 0 and the last digit), and the middle digit (5 choices):

$$3 \times 5 \times 5 = 75$$

(c) Numbers greater than 330 must either start with 3 and have a second digit > 3 , or start with > 3 .

Case 1: Start with 3. Second digit > 3 : {4, 5, 6} (3 choices). Third digit: 5 choices:

$$3 \times 5 = 15$$

Case 2: Start with > 3 : {4, 5, 6} (3 choices). Remaining digits: $6 \times 5 = 30$:

$$3 \times 30 = 90$$

Total:

$$15 + 90 = 105$$

Answers:

- (a)
(b)
(c)



2.37

Question: In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?

Solution:

The arrangement must alternate starting with either a boy or a girl.

Case 1: Start with a boy. Arrange the boys (4!) and the girls (5!):

$$4! \times 5! = 24 \times 120 = 2880$$

Case 2: Start with a girl. Similarly:

$$5! \times 4! = 120 \times 24 = 2880$$

Total:

$$2880 + 2880 = 5760$$

Answer:

2.38

Question: Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated:

- (a) with no restrictions?
- (b) if each couple is to sit together?
- (c) if all the men sit together to the right of all the women?

Solution:

- (a) No restrictions:

$$8! = 40,320$$

- (b) Treat each couple as a "block." Arrange the 4 blocks ($4!$) and arrange the individuals within each block ($2!$ per block):

$$4! \times (2!)^4 = 24 \times 16 = 384$$

- (c) Arrange the women ($4!$) and the men ($4!$):

$$4! \times 4! = 24 \times 24 = 576$$

Answers:

- (a)

40,320

- (b)

384

- (c)

576

2.39

Question: In a regional spelling bee, the 8 finalists consist of 3 boys and 5 girls. Find the number of sample points in the sample space S for the number of possible orders at the conclusion of the contest for:

- (a) all 8 finalists;
- (b) the first 3 positions.

Solution:

- (a) Total arrangements of 8 finalists:

$$8! = 40,320$$

- (b) Choose the first 3 positions from 8 finalists:

$$P(8, 3) = \frac{8!}{(8-3)!} = 8 \times 7 \times 6 = 336$$

Answers:

- (a)

40,320

 - (b)

336

-

2.40

Question: In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?

Solution:

Choose 5 players from 8 and arrange them:

$$P(8, 5) = \frac{8!}{(8-5)!} = 8 \times 7 \times 6 \times 5 \times 4 = 6720$$

Answer: 6720

2.41

Question: Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.

Solution:

Assign 4 teachers to the sections ($P(6, 4)$):

$$P(6, 4) = \frac{6!}{(6-4)!} = 6 \times 5 \times 4 \times 3 = 360$$

Answer: 360

2.42

Question: Three lottery tickets for first, second, and third prizes are drawn from a group of 40 tickets. Find the number of sample points in S for awarding the 3 prizes if each contestant holds only 1 ticket.

Solution:

Choose and arrange 3 winners from 40:

$$P(40, 3) = \frac{40!}{(40-3)!} = 40 \times 39 \times 38 = 59,280$$

Answer: 59,280

2.43

Question: In how many ways can 5 different trees be planted in a circle?

Solution:

In a circular arrangement, fix one tree and arrange the remaining 4:

$$(5-1)! = 4! = 24$$

Answer: 24

2.44

Question: In how many ways can a caravan of 8 covered wagons from Arizona be arranged in a circle?

Solution:

Fix one wagon and arrange the remaining 7:

$$(8 - 1)! = 7! = 5040$$

Answer:

2.45

Question: How many distinct permutations can be made from the letters of the word INFINITY?

Solution:

The word "INFINITY" has 8 letters, including 3 I's, 2 N's, and 1 each of F, T, Y. The total number of permutations is:

$$\frac{8!}{3! \times 2!} = \frac{40,320}{6 \times 2} = 3360$$

Answer:

2.46

Question: In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish among trees of the same kind?

Solution:

The total number of trees is $3 + 4 + 2 = 9$. The number of arrangements is:

$$\frac{9!}{3! \times 4! \times 2!} = \frac{362,880}{6 \times 24 \times 2} = 1260$$

Answer:

2.47

Question: How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?

Solution:

This is a combination problem:

$$\binom{8}{3} = \frac{8!}{3!(8-3)!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Answer:

2.48

Question: How many ways are there that no two students will have the same birth date in a class of size 60?

Solution:

There are 365 days in a year. The first student has 365 choices, the second has 364, and so on. The total number of ways is:

$$P(365, 60) = \frac{365!}{(365 - 60)!}$$

This value is extremely large and typically computed using software.

Answer: $P(365, 60)$

2.1 List the elements of each of the following sample spaces:

(a) the set of integers between 1 and 50 divisible by 8; 2

(b) the set $S = \{x \mid x^2 + 4x - 5 = 0\}$;

(c) the set of outcomes when a coin is tossed until a tail or three heads appear;

(d) the set $S = \{x \mid x \text{ is a continent}\}$;

(e) the set $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$

(a) The set of integers between 1 and 50 divisible by 8:

- The integers divisible by 8 are:
8, 16, 24, 32, 40, 48

Sample space:

$\{8, 16, 24, 32, 40, 48\}$

(b) The set $S = \{x \mid x^2 + 4x - 5 = 0\}$:

- Solve the quadratic equation:
 $x^2 + 4x - 5 = 0$
Factorizing:
 $(x + 5)(x - 1) = 0$
So, $x = -5$ or $x = 1$.

Sample space:

$\{-5, 1\}$

(c) The set of outcomes when a coin is tossed until a tail or three heads appear:

- 1. Tail appears on the first toss:**
 - The sequence is simply: **T** (we stop after the first tail).
- 2. First toss is Head, then Tail appears:**
 - The sequence is: **H T**.
- 3. First two tosses are Heads, then a Tail appears:**
 - The sequence is: **H H T**.
- 4. Three heads in a row (stopping after the third head, even if no tail appears):**
 - The sequence is: **H H H**.

: Sample Space

$$S = \{T, HT, HHT, HHH\}$$

(d) The set $S = \{x \mid x \text{ is a continent}\}$:

- The continents are:
Africa, Antarctica, Asia, Europe, North America, South America, Australia.

Sample space:

$\{\text{Africa, Antarctica, Asia, Europe, North America, South America, Australia}\}$

(e) The set $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$:

- Solve the inequalities:
 1. $2x - 4 \geq 0 \Rightarrow x \geq 2$
 2. But the second condition is $x < 1$.
- There is **no value** of x that satisfies both conditions.

Sample space:

$\emptyset$ (empty set)

2.2 Use the rule method to describe the sample space S consisting of all points in the first quadrant inside a circle of radius 3 with center at the origin.

To describe the sample space S using the **rule method**, we define it by specifying the conditions that elements (points) in the set must satisfy.

Given Conditions:

1. The circle has a **radius** of 3 and is centered at the **origin** $(0, 0)$.
 - The equation of the circle is:
$$x^2 + y^2 \leq 9$$
2. The points are located in the **first quadrant**, meaning both coordinates must be **non-negative**:
$$x \geq 0 \quad \text{and} \quad y \geq 0$$

Rule Method Description:

$$S = \{(x, y) \mid x^2 + y^2 \leq 9, x \geq 0, y \geq 0\}$$

2.3 Which of the following events are equal?

- (a) $A = \{1, 3\}$;
- (b) $B = \{x \mid x \text{ is a number on a die}\}$;
- (c) $C = \{x \mid x - 4x + 3 = 0\}$;
- (d) $D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}$.

(a) $A = \{1, 3\}$

This is a simple set containing the elements 1 and 3.

(b) $B = \{x \mid x \text{ is a number on a die}\}$

- A standard die has numbers from 1 to 6.
- So,

$$B = \{1, 2, 3, 4, 5, 6\}$$

(c) $C = \{x \mid x^2 - 4x + 3 = 0\}$

- Solve the quadratic equation:

$$x^2 - 4x + 3 = 0$$

Factorizing:

$$(x - 1)(x - 3) = 0$$

So, $x = 1$ or $x = 3$.

- Therefore,

$$C = \{1, 3\}$$

(d) $D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}$

- The number of heads when tossing 6 coins can range from 0 to 6.
- Therefore,

$$D = \{0, 1, 2, 3, 4, 5, 6\}$$

✓ Which Events Are Equal?

- $A = \{1, 3\}$ and $C = \{1, 3\} \rightarrow$ These two sets are equal.
- $B = \{1, 2, 3, 4, 5, 6\}$ and $D = \{0, 1, 2, 3, 4, 5, 6\} \rightarrow$ These are **not equal** due to the presence of 0 in D .

2.4 An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up. If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S

(a) by listing the elements (x, y) ;

(b) by using the rule method

✓ (a) Listing the Elements (x, y)

Each outcome is an ordered pair (x, y) , where x is the green die outcome and y is the red die outcome.

$$\begin{aligned} S = \{ & (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ & (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ & (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ & (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ & (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ & (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) \} \end{aligned}$$

- Total elements = $6 \times 6 = 36$.

✓ (b) Using the Rule Method

$$S = \{(x, y) \mid x \in \{1, 2, 3, 4, 5, 6\}, y \in \{1, 2, 3, 4, 5, 6\}\}$$

This means S includes **all ordered pairs** where:

- x is the outcome on the **green die**.
- y is the outcome on the **red die**.
- Both x and y can take any value from **1 to 6**.

2.5 An experiment consists of tossing a die and then flipping a coin once if the number on the die is even. If the number on the die is odd, the coin is flipped twice. Using the notation 4H, for example, to denote the outcome that the die comes up 4 and then the coin comes up heads, and 3HT to denote the outcome that the die comes up 3 followed by a head and then a tail on the coin, construct a tree diagram to show the 18 elements of the sample space S

Steps to Construct the Tree Diagram

1. Tossing the Die:

The die can land on 1, 2, 3, 4, 5, or 6.

2. Flipping the Coin:

- If the die shows an **even number** (2, 4, 6), flip the coin **once** → Outcomes: H or T.
- If the die shows an **odd number** (1, 3, 5), flip the coin **twice** → Outcomes: HH, HT, TH, or TT.

Sample Space (18 Elements)

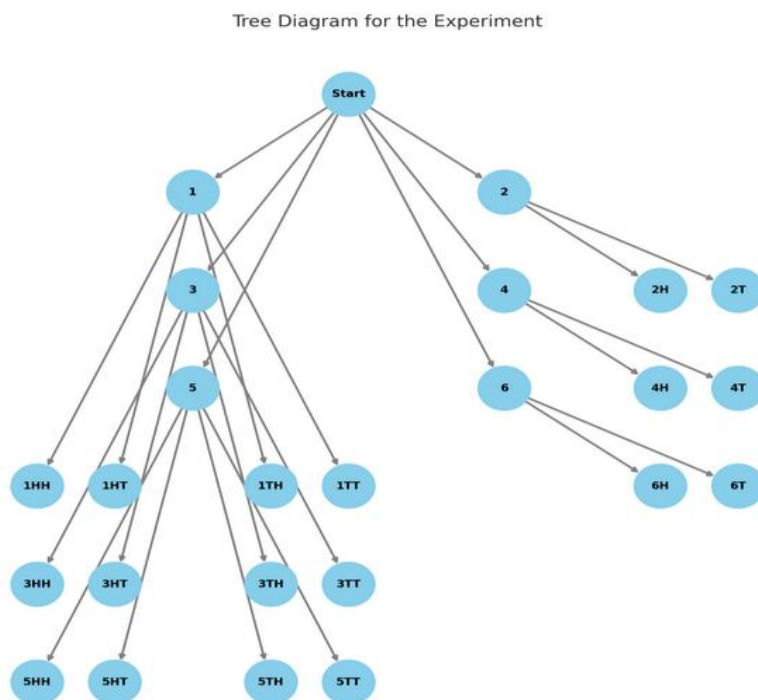
- For even die outcomes (each with 2 possibilities):
 - 2H, 2T
 - 4H, 4T
 - 6H, 6T
- For odd die outcomes (each with 4 possibilities):
 - 1HH, 1HT, 1TH, 1TT
 - 3HH, 3HT, 3TH, 3TT
 - 5HH, 5HT, 5TH, 5TT

Tree Diagram Description

1. **First Level (Die Roll):** Branches for 1, 2, 3, 4, 5, 6.

2. **Second Level (Coin Flip):**

- For **even** die results (2, 4, 6), each has two branches: **H** and **T**.
- For **odd** die results (1, 3, 5), each has four branches representing two coin flips:
 - First branch: **H** or **T** for the first flip.
 - Each of these branches splits again into **H** or **T** for the second flip.



2.6 Two jurors are selected from 4 alternates to serve at a murder trial. Using the notation $A_1 A_3$, for example, to denote the simple event that alternates 1 and 3 are selected, list the 6 elements of the sample space S .

To list the elements of the sample space S , let's break down the problem:

- There are 4 alternates, labeled **A1, A2, A3, A4**.
- Two jurors are selected, so we are choosing 2 out of the 4 alternates without regard to the order.

Step 1: List all possible pairs of alternates that could be selected.

Since order doesn't matter, we simply list the combinations of 2 alternates from the set $\{A1, A2, A3, A4\}$.

The possible combinations are:

1. $A1A2$
2. $A1A3$
3. $A1A4$
4. $A2A3$
5. $A2A4$
6. $A3A4$

Sample Space S :

$$S = \{A1A2, A1A3, A1A4, A2A3, A2A4, A3A4\}$$

These are the 6 possible outcomes of selecting two jurors from the four alternates.

Possible selections of 2 jurors from 4 alternates:

$$S = \{A_1A_2, A_1A_3, A_1A_4, A_2A_3, A_2A_4, A_3A_4\}.$$

2.7 Four students are selected at random from a chemistry class and classified as male or female. List the elements of the sample space S , using the letter M for male and F for female. Define a second sample space S_2 where the elements represent the number of females selected.

Sample Space S (4 students selected, classified as male or female)

Each student can either be male (M) or female (F). Since 4 students are selected, the total number of possible outcomes is $2^4=16$ because each student has two possibilities (M or F).

The sample space S contains all the possible outcomes for 4 students, where each outcome is represented by a combination of M's and F's.

The elements of the sample space S are:

$S=\{MMMM,$
 $MMFM,MMFF,MFMM,MFMF,MFFM,MFFF,FMMM,FMFM,FMFF,FMME,FFME,FFFF,FFMM,FFME,FFMF\}$

Second Sample Space S_2 (Number of females selected)

In this sample space, we focus on how many females were selected from the four students. The possible number of females selected can range from 0 (no females) to 4 (all females). So, the elements of the sample space S_2 will be the possible number of females in each outcome.

The sample space S_2 will contain the numbers:

$$S_2 = \{0, 1, 2, 3, 4\}$$

Where:

- 0 means no females were selected (all males),
- 1 means one female was selected,
- 2 means two females were selected,
- 3 means three females were selected,
- 4 means all four selected students are female.

2.8 For the sample space of Exercise 2.4,

(a) list the elements corresponding to the event A that the sum is greater than 8;

(b) list the elements corresponding to the event B that a 2 occurs on either die;

(c) list the elements corresponding to the event C that a number greater than 4 comes up on the green die;

(d) list the elements corresponding to the event $A \cap C$;

(e) list the elements corresponding to the event $A \cap B$;

(f) list the elements corresponding to the event $B \cap C$;

(g) construct a Venn diagram to illustrate the intersections and unions of the events A , B , and C .

Sample Space S (for tossing two dice)

The elements of the sample space S are:

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

(a) Event A (Sum greater than 8)

Event A corresponds to the outcomes where the sum of x and y (i.e., $x + y$) is greater than 8. We need to find the pairs where the sum exceeds 8:

The possible pairs are:

$$A = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

(b) Event B (A 2 occurs on either die)

Event B corresponds to the outcomes where at least one die shows a 2. So, we need to list all outcomes where either $x = 2$ or $y = 2$ (or both):

The possible pairs are:

$$B = \{(1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (4, 2), (5, 2), (6, 2)\}$$

(c) Event C (A number greater than 4 on the green die)

Event C corresponds to the outcomes where the number on the green die x is greater than 4. Thus, x can be 5 or 6, and we list all possible outcomes for y (red die):

The possible pairs are:

$$C = \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

(d) Event $A \cap C$ (Sum greater than 8 and number greater than 4 on the green die)

The intersection $A \cap C$ consists of the outcomes where both the sum is greater than 8 and the number on the green die is greater than 4. From the previous results, we take only the outcomes where x is 5 or 6, and the sum is greater than 8:

The possible pairs are:

$$A \cap C = \{(5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

(e) Event $A \cap B$ (Sum greater than 8 and a 2 occurs on either die)

The intersection $A \cap B$ consists of the outcomes where both the sum is greater than 8 and there is a 2 on either die. From the previous results, we find the pairs that satisfy both conditions:

The possible pairs are:

$$A \cap B = \{(2, 6), (3, 6), (4, 5), (5, 4), (5, 5), (6, 3)\}$$

(f) Event $B \cap C$ (A 2 occurs on either die and a number greater than 4 on the green die)

The intersection $B \cap C$ consists of the outcomes where either die shows a 2 and the number on the green die is greater than 4. From the previous results, we find the pairs that satisfy both conditions:

The possible pairs are:

$$B \cap C = \{(5, 2), (6, 2)\}$$

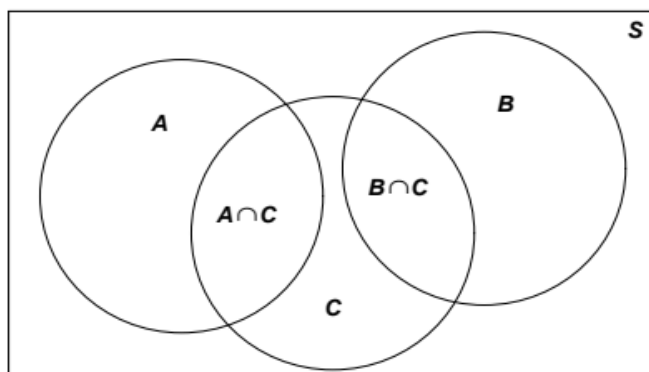


(g) Venn Diagram

In the Venn diagram, we would have three sets representing events A, B, and C:

- A represents outcomes where the sum is greater than 8.
- B represents outcomes where a 2 occurs on either die.
- C represents outcomes where the number on the green die is greater than 4

Venn Diagram of Events A, B, and C



Here is the Venn diagram illustrating the intersections and unions of the events A, B, and C:

- **Set A** represents outcomes where the sum is greater than 8.
- **Set B** represents outcomes where a 2 occurs on either die.
- **Set C** represents outcomes where the number on the green die is greater than 4.

2.9 For the sample space of Exercise 2.5,

(a) list the elements corresponding to the event A that a number less than 3 occurs on the die;

(b) list the elements corresponding to the event B that two tails occur;

(c) list the elements corresponding to the event A ;

(d) list the elements corresponding to the event $A \cap B$;

(e) list the elements corresponding to the event $A \cup B$.

Sample Space S :

$$S = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$$

(a) List the elements corresponding to the event A that a number less than 3 occurs on the die:

Event A corresponds to the outcomes where the die shows either 1 or 2.

- For 1 (odd): Coin is flipped twice $\rightarrow 1HH, 1HT, 1TH, 1TT$
- For 2 (even): Coin is flipped once $\rightarrow 2H, 2T$

Thus, the elements corresponding to event A are:

$$A = \{1HH, 1HT, 1TH, 1TT, 2H, 2T\}$$

(b) List the elements corresponding to the event B that two tails occur:

Event B corresponds to the outcomes where the coin shows two tails.

- 1 (odd): Coin flipped twice $\rightarrow 1TT$
- 3 (odd): Coin flipped twice $\rightarrow 3TT$
- 5 (odd): Coin flipped twice $\rightarrow 5TT$

Thus, the elements corresponding to event B are:

$$B = \{1TT, 3TT, 5TT\}$$

(c) List the elements corresponding to the event A' (the complement of event A , meaning the die shows a number greater than or equal to 3):

Event A' corresponds to the outcomes where the die shows 3, 4, 5, or 6.

- For 3 (odd): Coin flipped twice $\rightarrow 3HH, 3HT, 3TH, 3TT$
- For 4 (even): Coin flipped once $\rightarrow 4H, 4T$
- For 5 (odd): Coin flipped twice $\rightarrow 5HH, 5HT, 5TH, 5TT$
- For 6 (even): Coin flipped once $\rightarrow 6H, 6T$

Thus, the elements corresponding to event A' are:

$$A' = \{3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$$

(d) List the elements corresponding to the event $A \cap B$ (the intersection of events A and B , meaning the die shows a number less than 3 and two tails occur):

- The outcomes where the die shows less than 3 and the coin shows two tails:
 - For 1 (odd): Coin flipped twice $\rightarrow 1TT$
 - For 2 (even): Cannot have two tails because only one coin is flipped.

Thus, the elements corresponding to $A \cap B$ are:

$$A \cap B = \{1TT\}$$

(e) List the elements corresponding to the event $A \cup B$ (the union of events A and B , meaning the die shows a number less than 3 or two tails occur):

- From Event A (numbers less than 3): $\{1HH, 1HT, 1TH, 1TT, 2H, 2T\}$
- From Event B (two tails): $\{1TT, 3TT, 5TT\}$

The union of these events combines the unique outcomes from both sets:

$$A \cup B = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3TT, 5TT\}$$

2.10 An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

(a) List the elements of a sample space S , using the letters F for safe to fish and N for not safe to fish.

(b) List the elements of S corresponding to event E that at least two of the rivers are safe for fishing.

(c) Define an event that has as its elements the points $\{FFF, NFF, FFN, NFN\}$.

(a) List the elements of the sample space S

- Each river can either be **safe (F)** or **not safe (N)** for fishing.
- There are **three rivers**, so each river has two possibilities: F or N .
- The total number of outcomes is $2^3 = 8$.

Thus, the sample space S is:

$$S = \{FFF, FFN, FNF, FNN, NFF, NFN, NNF, NNN\}$$

(b) List the elements of S corresponding to event E that at least two of the rivers are safe for fishing.

- "At least two safe rivers" means the outcomes with exactly two or exactly three F's.

From the sample space, these outcomes are:

1. $FFF \rightarrow$ all three are safe.
2. $FFN \rightarrow$ first two are safe.
3. $FNF \rightarrow$ first and third are safe.
4. $NFF \rightarrow$ second and third are safe.

Thus, event E is:

$$E = \{FFF, FFN, FNF, NFF\}$$

(c) Define an event that has as its elements the points $\{FFF, NFF, FFN, NFN\}$.

Let's analyze the common feature of these outcomes:

- FFF : All rivers are safe.
- NFF : The first river is not safe, but the second and third are safe.
- FFN : The first two rivers are safe, but the third is not.
- NFN : Only the second river is safe.

Observation:

- The **second river** is safe (F) in **all four outcomes**.
- The first and third rivers can be either safe or not safe.

2.11 The resu´es of two male applicants for a college teaching position in chemistry are placed in the same file as the resu´es of two female applicants. Two po- sitions become available, and the first, at the rank of assistant professor, is filled by selecting one of the four applicants at random. The second position, at the rank of instructor, is then filled by selecting at random one of the remaining three applicants. Using the notation $M F$, for example, to denote the simple event that

the first position is filled by the second male applicant and the second position is then filled by the first female applicant,

- (a) list the elements of a sample space S ;
- (b) list the elements of S corresponding to event A that the position of assistant professor is filled by a male applicant;
- (c) list the elements of S corresponding to event B that exactly one of the two positions is filled by a male applicant;
- (d) list the elements of S corresponding to event C that neither position is filled by a male applicant;
- (e) list the elements of S corresponding to the event $A \cap B$;
- (f) list the elements of S corresponding to the event $A \cup C$;
- (g) construct a Venn diagram to illustrate the intersec- tions and unions of the events A , B , and C .

(a) List the elements of a sample space S :

The sample space consists of all possible ways to assign the two positions, starting with the assistant professor position and then the instructor position.

The possible combinations are:

1. **M1M2**: First position filled by M1, second position filled by M2.
2. **M1F1**: First position filled by M1, second position filled by F1.
3. **M1F2**: First position filled by M1, second position filled by F2.
4. **M2M1**: First position filled by M2, second position filled by M1.
5. **M2F1**: First position filled by M2, second position filled by F1.
6. **M2F2**: First position filled by M2, second position filled by F2.
7. **F1M1**: First position filled by F1, second position filled by M1.
8. **F1M2**: First position filled by F1, second position filled by M2.
9. **F1F2**: First position filled by F1, second position filled by F2.
10. **F2M1**: First position filled by F2, second position filled by M1.
11. **F2M2**: First position filled by F2, second position filled by M2.
12. **F2F1**: First position filled by F2, second position filled by F1.

Thus, the sample space S consists of 12 outcomes:

$$S = \{M1M2, M1F1, M1F2, M2M1, M2F1, M2F2, F1M1, F1M2, F1F2, F2M1, F2M2, F2F1\}$$

(b) List the elements of S corresponding to event A that the position of assistant professor is filled by a male applicant:

Event A is the event where the assistant professor position is filled by one of the two male applicants.

This means the first position must be filled by $M1$ or $M2$.

The relevant outcomes are:

- $M1M2$
- $M1F1$
- $M1F2$
- $M2M1$
- $M2F1$
- $M2F2$

Thus, the elements corresponding to event A are:

$$A = \{M1M2, M1F1, M1F2, M2M1, M2F1, M2F2\}$$

(c) List the elements of S corresponding to event B that exactly one of the two positions is filled by a male applicant:

Event B corresponds to the case where one male and one female are selected for the two positions. The outcomes where **exactly one position is filled by a male** are those where one male and one female are selected.

The relevant outcomes are:

- $M1F1$
- $M1F2$
- $M2F1$
- $M2F2$
- $F1M1$
- $F1M2$
- $F2M1$
- $F2M2$

Thus, the elements corresponding to event B are:

$$B = \{M1F1, M1F2, M2F1, M2F2, F1M1, F1M2, F2M1, F2M2\}$$

(d) List the elements of S corresponding to event C that neither position is filled by a male applicant:

Event C corresponds to the case where both positions are filled by **female applicants**. This means both positions must be filled by $F1$ and $F2$.

The relevant outcome is:

- $F1F2$
- $F2F1$

Thus, the elements corresponding to event C are:

$$C = \{F1F2, F2F1\}$$

(e) List the elements of S corresponding to the event $A \cap B$ (the intersection of events A and B , meaning exactly one position is filled by a male applicant and the assistant professor position is filled by a male applicant):

Event $A \cap B$ corresponds to the outcomes where:

- The assistant professor position is filled by a male (event A).
- Exactly one of the two positions is filled by a male (event B).

This means the first position must be filled by a male, and the second position must be filled by a female.

The relevant outcomes are:

- M1F1
- M1F2
- M2F1
- M2F2

Thus, the elements corresponding to $A \cap B$ are:

$$A \cap B = \{M1F1, M1F2, M2F1, M2F2\}$$

(f) List the elements of S corresponding to the event $A \cup C$ (the union of events A and C , meaning the assistant professor position is filled by a male applicant or neither position is filled by a male applicant):

Event $A \cup C$ corresponds to the outcomes where:

- The assistant professor position is filled by a male (event A), or
- Neither position is filled by a male (event C).

The relevant outcomes are:

- M1M2, M1F1, M1F2, M2M1, M2F1, M2F2 (from event A).
- F1F2, F2F1 (from event C).

Thus, the elements corresponding to $A \cup C$ are:

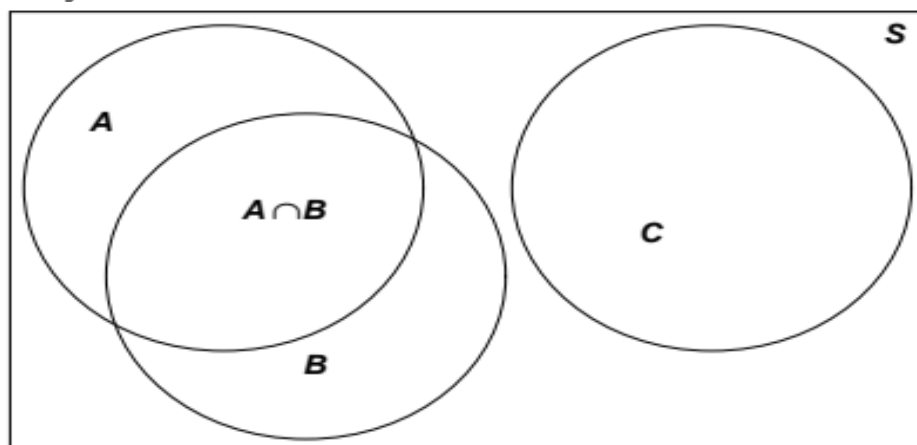
$$A \cup C = \{M1M2, M1F1, M1F2, M2M1, M2F1, M2F2, F1F2, F2F1\}$$

(g) Construct a Venn diagram to illustrate the intersections and unions of the events A , B , and C :

In a Venn diagram:

- A represents the set where the assistant professor position is filled by a male.
- B represents the set where exactly one position is filled by a male.
- C represents the set where neither position is filled by a male.

The diagram will show:



2.12 Exercise and diet are being studied as possible substitutes for medication to lower blood pressure. Three groups of subjects will be used to study the effect of exercise. Group 1 is sedentary, while group 2 walks and group 3 swims for 1 hour a day. Half of each of the three exercise groups will be on a salt-free diet. An additional group of subjects will not exercise or restrict their salt, but will take the standard medication. Use Z for sedentary, W for walker, S for swimmer, Y for salt, N for no salt, M for medication, and F for medication free.

(a) Show all of the elements of the sample space S .

(b) Given that A is the set of nonmedicated subjects and B is the set of walkers, list the elements of $A \cup B$.

(c) List the elements of $A \cap B$.

(a) Show all of the elements of the sample space S :

To form the sample space S , we need to list all possible combinations of exercise, diet, and medication status.

Groups with exercise and diet:

- Group 1 (Sedentary): Half of them on salt-free diet, and the other half on regular diet.
 - ZY (Sedentary and salt-free diet)
 - ZN (Sedentary and regular diet)
- Group 2 (Walker): Half of them on salt-free diet, and the other half on regular diet.
 - WY (Walker and salt-free diet)
 - WN (Walker and regular diet)
- Group 3 (Swimmer): Half of them on salt-free diet, and the other half on regular diet.
 - SY (Swimmer and salt-free diet)
 - SN (Swimmer and regular diet)

Medication group:

- Medication group (no exercise or diet):
 - M (Medication, no exercise or salt restriction)
 - F (Medication-free, no exercise or salt restriction)

Now, the sample space S is the set of all possible combinations of the exercise group, diet, and medication status:

$$S = \{ZY, ZN, WY, WN, SY, SN, M, F\}$$

(b) Given that A is the set of nonmedicated subjects and B is the set of walkers, list the elements of $A \cup B$:

- A is the set of nonmedicated subjects, which are those with medication-free (F) status.
 - $A = \{ZY, ZN, WY, WN, SY, SN, F\}$
- B is the set of walkers, which includes the outcomes where the exercise is walking (W).
 - $B = \{WY, WN\}$

The union of sets $A \cup B$ includes all elements from both A and B (without repetition):

$$A \cup B = \{ZY, ZN, WY, WN, SY, SN, F\}$$

(c) List the elements of $A \cap B$:

The intersection of A and B consists of the elements that are **both nonmedicated (F) and walkers (W)**.

- From $A = \{ZY, ZN, WY, WN, SY, SN, F\}$
- From $B = \{WY, WN\}$

The common elements between A and B are:

- WY (Walker and salt-free diet, nonmedicated)
- WN (Walker and regular diet, nonmedicated)

Thus, the intersection $A \cap B$ is:

$$A \cap B = \{WY, WN\}$$

2.13 Construct a Venn diagram to illustrate the possible intersections and unions for the following events relative to the sample space consisting of all automobiles made in the United States.

F : Four door, S : Sun roof, P : Power steering.

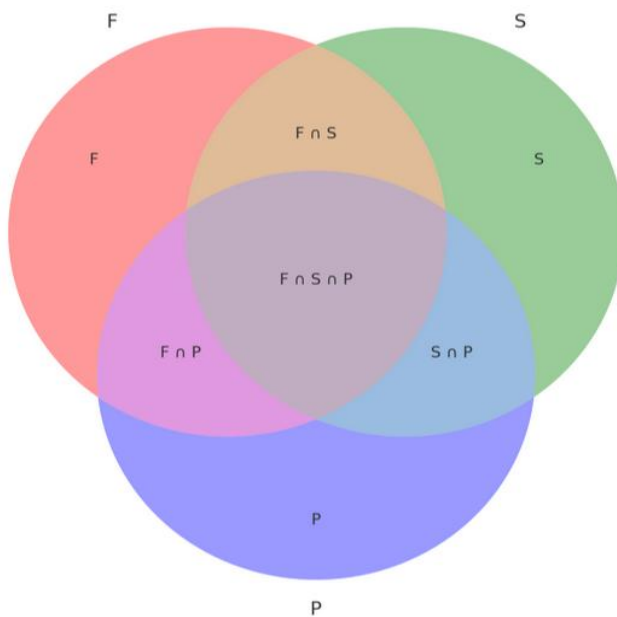
To create a Venn diagram illustrating the possible intersections and unions of the given events relative to the sample space of all automobiles made in the United States, we need to represent:

- **F:** Cars with four doors
- **S:** Cars with a sunroof
- **P:** Cars with power steering

Steps for the Venn Diagram:

1. Draw three overlapping circles, each representing one of the events (F, S, and P).
2. Label each circle appropriately.
3. The **union** of two or more sets represents cars that have at least one of those features.
4. The **intersection** of two or more sets represents cars that have all those features

Venn Diagram for Automobiles Made in the U.S.



Here is the Venn diagram for the events related to automobiles made in the U.S.:

- **F** represents four-door cars.
- **S** represents cars with a sunroof.
- **P** represents cars with power steering.

2.14 Given the sets:

- $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $A = \{0, 2, 4, 6, 8\}$
- $B = \{1, 3, 5, 7, 9\}$
- $C = \{2, 3, 4, 5\}$
- $D = \{1, 6, 7\}$

List the elements of the sets corresponding to the following events:

- $A \cup C$
- $A \cap B$
- C'
- $(C' \cap D) \cup B$
- $(S \cap C')'$
- $A \cap C \cap D'$

(a) $A \cup C$ (Union)

$$A \cup C = \{0, 2, 4, 6, 8\} \cup \{2, 3, 4, 5\} = \{0, 2, 3, 4, 5, 6, 8\}$$

(b) $A \cap B$ (Intersection)

$$A \cap B = \{0, 2, 4, 6, 8\} \cap \{1, 3, 5, 7, 9\} = \emptyset$$

(No common elements)

(c) C' (Complement of C in S)

$$C' = S - C = \{0, 1, 6, 7, 8, 9\}$$

(d) $(C' \cap D) \cup B$

First, find $C' \cap D$:

$$C' \cap D = \{0, 1, 6, 7, 8, 9\} \cap \{1, 6, 7\} = \{1, 6, 7\}$$

Now, take the union with B:

$$(C' \cap D) \cup B = \{1, 6, 7\} \cup \{1, 3, 5, 7, 9\} = \{1, 3, 5, 6, 7, 9\}$$

(e) $(S \cap C)'$

Since $S \cap C = C$, we have:

$$(S \cap C)' = (C)' = C' = \{0, 1, 6, 7, 8, 9\}$$

(f) $A \cap C \cap D'$

First, find $A \cap C$:

$$A \cap C = \{0, 2, 4, 6, 8\} \cap \{2, 3, 4, 5\} = \{2, 4\}$$

Next, find D' (Complement of D in S):

$$D' = S - D = \{0, 2, 3, 4, 5, 8, 9\}$$

Now, intersect with $A \cap C$:

$$A \cap C \cap D' = \{2, 4\} \cap \{0, 2, 3, 4, 5, 8, 9\} = \{2, 4\}$$

2.15 Consider the sample space:

- $S = \{\text{copper, sodium, nitrogen, potassium, uranium, oxygen, zinc}\}$

And the events:

- $A = \{\text{copper, sodium, zinc}\}$
- $B = \{\text{sodium, nitrogen, potassium}\}$
- $C = \{\text{oxygen}\}$

List the elements of the sets corresponding to the following events:

(a) A'

(b) $A \cup C$

(c) $(A \cap B') \cup C'$

(d) $B' \cap C'$

(e) $A \cap B \cap C$

(f) $(A' \cup B') \cap (A' \cap C)$

(a) A' (Complement of A in S)

$$A' = S - A = \{\text{nitrogen, potassium, uranium, oxygen}\}$$

(b) $A \cup C$ (Union)

$$A \cup C = \{\text{copper, sodium, zinc}\} \cup \{\text{oxygen}\} = \{\text{copper, sodium, zinc, oxygen}\}$$

(c) $(A \cap B') \cup C'$

First, find B' (Complement of B in S):

$$B' = S - B = \{\text{copper, uranium, oxygen, zinc}\}$$

Now, find $A \cap B'$:

$$A \cap B' = \{\text{copper, sodium, zinc}\} \cap \{\text{copper, uranium, oxygen, zinc}\} = \{\text{copper, zinc}\}$$

Now, find C' (Complement of C in S):

$$C' = S - C = \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\}$$

Now, take the union:

$$(A \cap B') \cup C' = \{\text{copper, zinc}\} \cup \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\} = \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\}$$

(d) $B' \cap C'$

$$B' \cap C' = \{\text{copper, uranium, oxygen, zinc}\} \cap \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\} = \{\text{copper, uranium, zinc}\}$$

(e) $A \cap B \cap C$

$$A \cap B \cap C = \{\text{copper, sodium, zinc}\} \cap \{\text{sodium, nitrogen, potassium}\} \cap \{\text{oxygen}\} = \emptyset$$

(No common elements)

(f) $(A' \cup B') \cap (A' \cap C)$

First, find $A' \cup B'$:

$$A' \cup B' = \{\text{nitrogen, potassium, uranium, oxygen}\} \cup \{\text{copper, uranium, oxygen, zinc}\} = \{\text{copper, nitrogen, potassium, uranium, oxygen, zinc}\}$$

Next, find $A' \cap C$:

$$A' \cap C = \{\text{nitrogen, potassium, uranium, oxygen}\} \cap \{\text{oxygen}\} = \{\text{oxygen}\}$$

Now, intersect both:

$$(A' \cup B') \cap (A' \cap C) = \{\text{copper, nitrogen, potassium, uranium, oxygen, zinc}\} \cap \{\text{oxygen}\} = \{\text{oxygen}\}$$

2.16 If $S = \{x \mid 0 < x < 12\}$, $M = \{x \mid 1 < x < 9\}$, and $N = \{x \mid 0 < x < 5\}$, find

(a) $M \cup N$;

(b) $M \cap N$;

(c) $M' \cap N'$.

Given Sets:

- $S = \{x \mid 0 < x < 12\} \rightarrow S = (0, 12)$
- $M = \{x \mid 1 < x < 9\} \rightarrow M = (1, 9)$
- $N = \{x \mid 0 < x < 5\} \rightarrow N = (0, 5)$

(a) $M \cup N$ (Union)

The union includes all elements in either M or N :

$$M \cup N = (1, 9) \cup (0, 5)$$

Since $(0, 5)$ and $(1, 9)$ overlap in $(1, 5)$, the complete union covers:

$$M \cup N = (0, 9)$$

(b) $M \cap N$ (Intersection)

The intersection includes only elements that are in both M and N :

$$M \cap N = (1, 9) \cap (0, 5)$$

The common portion between $(1, 9)$ and $(0, 5)$ is:

$$M \cap N = (1, 5)$$

(c) $M' \cap N'$ (Complement Intersection)

First, find the complements:

- $M' = S - M = (0, 1] \cup [9, 12)$
- $N' = S - N = [5, 12)$

Now, find the intersection:

$$M' \cap N' = [(0, 1] \cup [9, 12)] \cap [5, 12)$$

The only overlapping region is:

$$M' \cap N' = [9, 12)$$

2.17 Let A , B , and C be events relative to the sample space S . Using Venn diagrams, shade the areas representing the following events:

- (a) $(A \cap B)'$;
- (b) $(A \cup B)'$;
- (c) $(A \cap C) \cup B$.

(a) $(A \cap B)'$ (Complement of Intersection)

- $A \cap B$ is the region where both A and B overlap.
- $(A \cap B)'$ is everything **outside** this intersection, which includes:
 - The area in A but not in B
 - The area in B but not in A
 - The area outside both A and B

Shaded Region: Everything except the overlap of A and B .

(b) $(A \cup B)'$ (Complement of Union)

- $A \cup B$ is the area covered by either A or B (or both).
- $(A \cup B)'$ is the area **outside** both A and B .

Shaded Region: The area outside $A \cup B$ (everything except A and B).

(c) $(A \cap C) \cup B$

- $A \cap C$ is where A and C overlap.
- $(A \cap C) \cup B$ includes everything in B as well as everything in the $A \cap C$ region.

Shaded Region: The entire B region and the intersection of A and C .

2.18 Which of the following pairs of events are mutually exclusive?

- (a) A golfer scoring the lowest 18-hole round in a 72-hole tournament and losing the tournament.
- (b) A poker player getting a flush (all cards in the same suit) and 3 of a kind on the same 5-card hand.
- (c) A mother giving birth to a baby girl and a set of twin daughters on the same day.
- (d) A chess player losing the last game and winning the match

(a) *A golfer scoring the lowest 18-hole round in a 72-hole tournament and losing the tournament.*

- These events are **not mutually exclusive**. A golfer can have the best score in one 18-hole round but still lose the overall 72-hole tournament if their performance in the other rounds is worse than others.
-

(b) *A poker player getting a flush (all cards in the same suit) and 3 of a kind on the same 5-card hand.*

- These events are **mutually exclusive**. A flush requires all 5 cards to be of the same suit, while 3 of a kind requires three cards of the same rank, and the suits don't matter. It's impossible for a 5-card hand to be both a flush and 3 of a kind at the same time.
-

(c) *A mother giving birth to a baby girl and a set of twin daughters on the same day.*

- These events are **not mutually exclusive**. The mother could give birth to triplets: one single girl and a set of twin daughters, making both events possible on the same day.
-

(d) *A chess player losing the last game and winning the match.*

- These events are **not mutually exclusive**. A player can win enough games earlier in the match to secure victory, even if they lose the last game.
-

✓ **Correct Answer: (b)**

